MI-23: Resolution

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Repository status: Solved. The complete argument received an independent Codex-agent PASS review against the exact catalog target. The original draft was AI-assisted; the reviewed proof below is unchanged. This is independent agent verification, not external human peer review or formal proof certification. Detailed review and scope.

1 MI-23: an exact counterexample to the generalized geometric-mean conjecture

For positive definite matrices, write

$$A\#_{r,t}B = A^{r/2}(A^{-1/2}BA^{-1/2})^t A^{r/2}.$$

Conjecture 2.1 of [1], reproduced as MI-23, asserts

$$\lambda\big((A\#_{r,t}B)^p(A\#_{s,1-t}B)^p\big) \prec_{\log} \lambda\big(A^{p(r+s-1)}B^p\big) \tag{1}$$

for $0 \le t \le 1$, $p \ge 1$, and either $r, s \ge 1$ or $r, s \le 0$. Products of two positive definite matrices have positive eigenvalues, which are decreasingly ordered here. The assertion fails already for $r = s = 1$ and $p = 2$.

Theorem 1.1. *Let*

$$D = \operatorname{diag}\left(16, \frac{1}{12}, 1\right), \qquad T = \begin{pmatrix} 2 & 1 & 2 \\ 1 & 25 & -10 \\ 2 & -10 & 10 \end{pmatrix}, \qquad A = D^2, \qquad B = DT^8D.$$

Then $A, B > 0$, and (1) is false at

$$r = s = 1, \qquad p = 2, \qquad t = \frac{1}{8}.$$

Proof. The leading principal minors of T are 2, 49, and 150. Thus $T > 0$, and consequently $A, B > 0$. Moreover,

$$A^{-1/2}BA^{-1/2} = T^8.$$

Its positive eighth root is exactly T, not merely a numerical approximation. Hence the two geometric means are the rational matrices

$$G = A\#_{1,1/8}B = DTD, \qquad H = A\#_{1,7/8}B = DT^7D.$$

The first inequality in the proposed log-majorization would imply

$$\lambda_{\max}(G^2H^2) \le \lambda_{\max}(A^2B^2).$$

Since G, H, A, B are positive definite,

$$\lambda_{\max}(G^2H^2) = \|GH\|_\infty^2, \qquad \lambda_{\max}(A^2B^2) = \|AB\|_\infty^2.$$

For example, G^2H^2 is similar to $HG^2H = (GH)^*(GH)$, which proves the first identity; the second is identical.

Exact rational matrix multiplication gives

$$(GH)_{13} = \frac{1260589125202}{9} > 140000000000,$$
$$\|AB\|_F^2 = \frac{2009446159144992718181231562721}{107495424} < (138000000000)^2.$$

It follows that

$$\|GH\|_\infty \geq |(GH)_{13}| > 140000000000 > 138000000000 > \|AB\|_F \geq \|AB\|_\infty,$$

contradicting the required largest-eigenvalue inequality. For an alternative one-line rational certificate of the strict comparison,

$$((GH)_{13})^2 - \|AB\|_F^2 = \frac{99434824489435745411095588895}{107495424} > 0.$$

□

Remark 1.2 (Exactness and parameter scope). All input matrices and every displayed certificate are rational. The representation $B = DT^8D$ eliminates all fractional-power computations from the proof. This counterexample is outside the narrower interval $1/4 \leq t \leq 3/4$ established in [1] for $p = 2$; it does not contradict that theorem. It disproves the stated conjecture for the full interval $0 \leq t \leq 1$.

References

[1] Mohammad M. Ghabries, Hassane Abbas, Bassam Mourad, and Abdallah Assi. New log-majorization results concerning eigenvalues and singular values and a complement of a norm inequality. arXiv:2105.13356, 2021. Conjectures 1.1 and 2.1. URL: https://arxiv.org/abs/2105.13356.